

Georgia Zachou

MSc Student in Astrophysics

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About Me

MSc Astrophysics student at LMU with experience in computational physics, scientific software development, and data analysis. Interested in **Bayesian inference**, and **AI applications** in cosmology.

Education

2025 – Now **MSc in Astrophysics**

University of Munich (LMU)

2019 – 2025 **BSc with Integrated MSc**

National Technical University of Athens

School of Applied Mathematical and Physical Sciences. GPA: 9.02/10 (Excellent)

Experience

Nov 2025 – Now **Power Electronics Team Member**

TUfast eco team, TUM

Designing and upgrading the inverter PCB for a solar racing car using Altium Designer, along with LTSpice for simulations and STM32CubeMX for microcontroller configuration.

Jul – Aug 2024 **Intern, Theoretical High Energy Physics Dept.**

NCSR “Demokritos”

Worked on Standard Model Effective Field Theory (SMEFT) using HELAC interface. Developed a Python library for symbolic calculations within this framework, demonstrated through Jupyter notebooks on [GitHub](#).

Feb 2023 – Mar 2024 **Technical Student**

TE-MPE-PE, CERN

Worked on the Muon Collider project as part of the [STEAM](#) team.

- Developed a C++ unified material-properties library with MatLab/Python wrappers, compatible with COMSOL, FiQuS, LEDET, and BBQ.
- Implemented validation pipelines and unit tests for numerical functions.
- Built and automated documentation pipelines for a multi-tool scientific software ecosystem using MkDocs, Jinja, and GitLab CI/CD ([link](#)).
- Automated compilation, testing and deployment using GitLab CI/CD.

Jan – Oct 2022 **Avionics & Payload Team Member**

White Noise Rocketry Team, NTUA

- Designed payload PCBs and sensor interfaces using KiCad.
- Developed low-level C drivers for accelerometer-based flight instrumentation.

Publications

2023 [An Open-Source 3D FE Quench Simulation Tool for No-Insulation HTS Pancake Coils.](#)

Awards

2019 – 2021 “E. Anastasaki & S.D.P. Vlassopoulou” Award of Excellence

2025 – 2026 “Syn-enosis” Scholarship/award for post-graduate studies

Skills

Programming	Python, MATLAB, HTML/CSS/JavaScript, C/C++, FORTRAN
Libraries & Data	NumPy, Matplotlib, Pandas, Jinja2, PyTorch, Keras, Astropy, emcee
Tools	GitLab, KiCad, Altium, LTSpice, LaTeX, MS Office
Languages	Greek (native), English (C2), French (B2), German (A2 – still learning)

Other Activities

Aug–Sep 2024 **SUMTRAIC**, Institute for Plasma Physics, Czech Republic. **Analyzed & presented spectroscopic measurements** during the transport barrier formation on GOLEM tokamak ([report](#)).

Apr 2024 Spring School of Gravity and Cosmology, National Observatory of Athens

Sep–Oct 2024 [2nd EPS TIG Hands-on Event](#), Technology & Interfaces, CERN

Jul 2022 [57th Summer School](#), National Centre for Scientific Research “Demokritos”

Apr 2022 QGreece Workshops QSilver